

Sharon Zhang

CONTACT

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INTERESTS

Content creation tools, LLM-based design tools, visual content verification, computational design, programmatic generation.

EDUCATION

2021 - Current **Stanford University**
PhD in Computer Science
Advisor: Maneesh Agrawala

2017 - 2021 **Princeton University**
A.B. in Mathematics with honors · Minor in Computer Science
Thesis: *Contextual Bias and Interpretability in Visual Classification*
Advisor: Olga Russakovsky

RESEARCH EXPERIENCE

2020 - 2021 **Visual AI Lab, Princeton University**
Advisor: Olga Russakovsky

2020 **Department of Mathematics, Princeton University**
Advisor: Amit Singer

2019 **SMALL REU, Williams College**
Advisor: Susan Loepp
Undergraduate researcher in the 2019 Commutative Algebra cohort.

INDUSTRY EXPERIENCE

2025 **Research Intern, Roblox**
Contributed to research and development on the Foundation AI team for LLM-assisted functional object and scene generation. Developed verification pipelines for evaluating generation results and populating knowledge bases for in-context examples.

2024 **Research Intern, Roblox**
Designed and implemented a procedural generation pipeline for creating obstacle course games as part of a larger project on training RL agents for helping assess obstacle course game difficulty.

2021 **Research Intern, Google**
Worked on layered video representations with the Visual Dynamics team. Hosted by Jonathan Huang and Vivek Rathod.

2020 **Software Engineering Intern, Google**
Hosted by the Android Camera Machine Intelligence team. Designed and implemented

an open-sourced API for Android developers to incorporate Google motion photos.

PUBLICATIONS

Sharon Zhang, Jiaju Ma, Jiajun Wu, Daniel Ritchie, Maneesh Agrawala. "Editing Motion Graphics Video via Motion Vectorization and Transformation." *SIGGRAPH Asia*, 2023.

Chen Geng*, Hong-Xing Yu*, **Sharon Zhang**, Maneesh Agrawala, Jiajun Wu. "Tree-Structured Shading Decomposition." *International Conference on Computer Vision (ICCV)*, 2023.

Sunnie S. Y. Kim, **Sharon Zhang**, Nicole Meister, Olga Russakovsky. "[Re] Don't Judge an Object By It's Context: Overcoming Contextual Bias." *Machine Learning Reproducibility Challenge 2020*. In *ReScience C Journal*.

Sharon Zhang, Amit Moscovich, Amit Singer. "Product Manifold Learning." *Artificial Intelligence and Statistics (AISTATS)*, 2021.

Erica Barrett*, Emil Graf*, Kimball Strong*, **Sharon Zhang***, S. Loepp. "Cardinalities of Prime Spectra of Precompletions." *AMS Contemporary Mathematics: "Commutative Algebra: 150 Years with Roger and Sylvia Wiegand."* **773** (2021), 133–152.

Erica Barrett*, Emil Graf*, Kimball Strong*, **Sharon Zhang***, S. Loepp. "Structure of spectra of precompletions." *Rocky Mountain J. Math.* **50** (2020), no. 6, 1965–1988.

ONGOING WORK

My current work covers verification for 2D/3D layout generation and RL-based tools for 3D game design. Feel free to reach out for more details.

AWARDS

2023 - 2026	NSF Graduate Research Fellowship
2023	Ford Foundation Fellowship Honorable Mention
2023	Paul & Daisy Soros Fellowship Finalist
2022, 2023, 2025	Brown Institute for Media Innovation Magic Grant
2021	Middleton Miller '29 Prize, <i>awarded for best independent work in mathematics</i>
2021	NSF Graduate Research Fellowship Honorable Mention

ACTIVITIES & SERVICE

	Reviewer for AISTATS, Eurographics, CHI, DIS
2023	Organizer, Stanford Graphics Café
2020	Teaching Assistant, Princeton University COS 429: Computer Vision
2019	Peer Tutor, Princeton University MAT 215: Honors Analysis
2019	Peer Tutor, Princeton University COS 126: Introduction to Computer Science

COMPUTER SKILLS

Languages	Python, Java, HTML/CSS, C/C++, Lua, bash shell scripting
Software & Tools	JAX, PyTorch, TensorFlow, Git, CVX, L ^A T _E X, Google Cloud, Adobe Creative Suite

March 31, 2026